DO NOT WRITE IN THIS AREA

12 $(ax^6)^{\frac{1}{n}} = 7x^3$

Work out the value of a and the value of n.

1 mark
compare powers 6/n=3

1 mark
a=49; n=2

$(ax^6)^{(1/n)} = a^{(1/n)}\, x^{(6/n)} = 7x^3$

$6/n = 3$, so $n = 2$

$\sqrt{a} = 7$, so $a = 49$

$a =$ 49

$n =$ 2

(Total for Question 12 is 2 marks)

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